

Catecholamines

Norepinephrine; ADHD

1. Where in the brainstem are the cell bodies of neurons that supply nearly all of the CNS with norepinephrine? Where else are NE neurons found?
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2. Identify the signal transduction pathway(s) of the following adrenergic receptors (AR).
 - α 1-AR
 - α 2-AR
 - β -AR
3. Describe the effects of the following substances on the delayed response task. Does adrenergic input to the PFC improve or impair working memory? Both? How?
 - norepinephrine
 - clonidine
 - How would you expect phenylephrine (α 1-AR agonist) or yohimbine (α 2-AR antagonist) to affect behavior on the delayed response task if each were administered to the prefrontal cortex?
4. What is ADHD?
 - a. What are the subtypes of ADHD?
 - b. What drug classes are used to treat ADHD? What is the basic mechanism of action of each class?
 - c. What are some of the concerns that have been raised with long-term treatment of children with psychostimulants?